

## Worksheet No. - 1

**Student Name:** Ankush kumar

**Branch:** MCA

**Semester:** 2<sup>nd</sup>

**Subject Name:** TECHNICAL TRAINING

**UID:** 25MCA20305

**Section/Group:** 1A

**Date of Performance:** 6<sup>th</sup> Jan, 2026

**Subject Code:** 25CAP-652

### Aim/Overview of the practical:

To design and implement a sample database system using DDL, DML, and DCL commands, including database creation, data manipulation, schema modification, and role-based access control to ensure data integrity and secure, read-only access for authorized users.

### Objective:

To gain practical experience in implementing Data Definition Language (DDL), Data Manipulation Language (DML), and Data Control Language (DCL) operations in a real database environment. This will also include implementing role-based privileges to secure data.

### Input/Apparatus Used:

- PostgreSQL
- pgAdmin

### Procedure/Algorithm/Code :

```
CREATE TABLE department (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(100) NOT NULL UNIQUE,  
    location VARCHAR(100)  
);  
  
CREATE TABLE employee (  
    emp_id INT PRIMARY KEY,
```

```
emp_name VARCHAR(100) NOT NULL,  
  
email VARCHAR(100) UNIQUE,  
  
salary INT CHECK (salary > 0),  
  
dept_id INT REFERENCES department(dept_id)  
  
);  
  
CREATE TABLE project (  
  
    project_id INT PRIMARY KEY,  
  
    project_name VARCHAR(100) NOT NULL,  
  
    emp_id INT REFERENCES employee(emp_id),  
  
    start_date DATE,  
  
    end_date DATE  
  
);  
  
INSERT INTO department VALUES  
  
(1, 'Engineering', 'Bangalore'),  
  
(2, 'HR', 'Delhi');  
  
INSERT INTO employee VALUES  
  
(101, 'Amit Kumar', 'amit@gmail.com', 65000, 1),  
  
(102, 'Neha Sharma', 'neha@gmail.com', 50000, 2);  
  
-- PROJECT ASSIGNMENT  
  
INSERT INTO project(project_id, project_name, emp_id, start_date, end_date)  
VALUES  
  
(1001, 'Employee Management System', 101, '2025-01-10', NULL);
```

```

SELECT * FROM department;

SELECT * FROM employee;

SELECT * FROM project;

-- UPDATE EMPLOYEE SALARY

UPDATE employee

SET salary = salary + 5000

WHERE emp_id = 101;

-- DELETE DEPARTMENT

DELETE FROM department WHERE dept_id = 2;

CREATE ROLE app_user

WITH LOGIN PASSWORD 'ankudata';

GRANT SELECT ON department, employee, project TO app_user;
  
```

### Output:

|   | dept_id<br>[PK] integer | dept_name<br>character varying (100) | location<br>character varying (100) |
|---|-------------------------|--------------------------------------|-------------------------------------|
| 1 | 1                       | Engineering                          | Bangalore                           |
| 2 | 2                       | HR                                   | Delhi                               |

|   | emp_id<br>[PK] integer | emp_name<br>character varying (100) | email<br>character varying (100) | salary<br>integer | dept_id<br>integer |
|---|------------------------|-------------------------------------|----------------------------------|-------------------|--------------------|
| 1 | 102                    | Neha Sharma                         | neha@gmail.com                   | 50000             | 2                  |
| 2 | 101                    | Amit Kumar                          | amit@gmail.com                   | 65000             | 1                  |

```

CREATE ROLE
Query returned successfully in 78 msec.
  
```

|   | project_id<br>[PK] integer | project_name<br>character varying (100) | emp_id<br>integer | start_date<br>date | end_date<br>date |
|---|----------------------------|-----------------------------------------|-------------------|--------------------|------------------|
| 1 | 1001                       | Employee Management Syst...             | 101               | 2025-01-10         | [null]           |

```

GRANT
Query returned successfully in 55 msec.
  
```

### **Learning outcomes (What I have learnt):**

1. I learned how to create multiple related tables using primary keys and foreign keys, which helped me understand how data in different tables can be connected logically.
2. I understood the importance of constraints like NOT NULL, UNIQUE, CHECK, and PRIMARY KEY to maintain data accuracy and prevent invalid entries.
3. I practiced all basic CRUD operations—inserting, retrieving, updating, and deleting records—which gave me a clear idea of how real-world databases are managed.
4. I implemented real-life logic such as reducing the number of available book copies when a book is issued, which helped me understand how business rules are applied in databases.
5. I learned how to create a database role and assign permissions, which made me understand the basics of database security and access control.